

HYDROGENATED NBR PATENTS: SYNTHESIS, HYDROGENATION CONTROL, AND FORMULATION TRENDS

1. ABSTRACT

This report synthesizes a patent landscape centered on hydrogenated nitrile butadiene rubber (HNBR) and closely related hydrogenated nitrile rubber materials, with emphasis on precursor composition, hydrogenation severity, catalyst choice, and downstream compounding. The target material is defined as NBR-derived copolymers in which a substantial fraction of butadiene unsaturation is hydrogenated to improve heat, ozone, oil, and fuel resistance while retaining processability and mechanical integrity. Across the selected patents, the dominant technical themes are: (i) direct hydrogenation of NBR in solution or latex using noble-metal catalysts, (ii) simultaneous hydrogenation/metathesis routes to reduce molecular weight and narrow molecular-weight distribution, (iii) control of residual unsaturation to balance cure response and aging, and (iv) formulation of HNBR into vulcanizates, hoses, seals, and nanocomposites for demanding automotive and oilfield uses.

The landscape includes early application-driven disclosures such as fuel-hose internal layers based on partially hydrogenated NBR with at least 50% hydrogenation of diene units, and later process patents focused on catalyst efficiency and catalyst-residue management. One patent discloses low-molecular-weight HNBR produced by simultaneous hydrogenation and metathesis using ruthenium/osmium catalysts, with catalyst loading, hydrogen pressure, and reaction time explicitly controlled. Another patent targets hydrogenation in the presence of Ru, Pd, or Rh compounds and specifies catalyst ppm ranges, temperature, pressure, and time to improve aging properties while avoiding post-hydrogenation catalyst-removal steps. Formulation-oriented patents show that HNBR performance can be tuned with sulfur/tellurium vulcanization systems, PVC or liquid NBR blending, reactive silicone oils, and functionalized fillers. Overall, the evidence indicates that the most important synthesis variables for HNBR are hydrogenation degree, catalyst identity and loading, solvent-phase processing, and post-reaction purification or compounding strategy.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US4486480A
- Patent Title: Rubber hose comprising a liquid nitrile-butadiene copolymer incorporated into a ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1984
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a fuel-hose construction whose internal layer is made from vulcanized partially hydrogenated unsaturated nitrile-conjugated diene copolymer rubber, i.e. hydrogenated NBR. The precursor rubber is described as being produced by emulsion polymerization or solution polymerization and then hydrogenated so that at least 50% of the conjugated diene units are saturated. - Hydrogenation degree: at least 50% of conjugated diene units — hydrogenated NBR definition - Hydrogenation degree: less than 98% desirable for proper cure rate — hydrogenated NBR definition - Unsaturated nitrile content: 10 to 70 wt.% — hydrogenated NBR definition - Liquid NBR number-average molecular weight: 500 to 10,000 — composition (C) - Liquid NBR preferred number-average molecular weight: 1000 to 5000 — composition (C) - Liquid NBR unsaturated nitrile content: 10 to 70 wt.% — composition (C) - Liquid NBR conjugated diene content: 20 to 80 wt.% — composition (C) - Liquid NBR vinyl monomer content: 0 to 20 wt.% — composition (C) - PVC incorporation: greater than 5 wt.% — composition (B) - PVC preferred incorporation: about 20 wt.% — composition (B) - PVC upper limit: 60 wt.% — composition (B) - Liquid NBR incorporation: greater than 2 wt.% — composition (C) - Liquid NBR upper limit: about 50 wt.% — composition (C)

Relevant Experimental Evidence - Example 1: hydrogenated NBR alone is used as the internal layer material. - Examples 2 to 6: PVC-containing hydrogenated NBR compounds improve dry properties and gasoline resistance versus hydrogenated NBR alone. - Examples 7 to 10: hydrogenated NBR-based internal layers show superior deteriorated-gasoline resistance, crack-propagation resistance, low-temperature resistance, and heat resistance.

Technical Relevance This patent is directly relevant because it defines HNBR by hydrogenation level and precursor composition, then demonstrates how HNBR is used in fuel-hose layers. It also shows practical formulation windows for blending PVC and liquid NBR to tune performance.

Patent 2

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight hydrogenated nitrile rubber prepared by simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and a ruthenium- or osmium-based catalyst. The process is designed to reduce molecular weight and narrow molecular-weight distribution while maintaining low residual double-bond content. - Hydrogenation/metathesis catalyst metal: ruthenium or osmium — claim 1 - Hydrogen pressure: 6.9×10^5 Pa — Example 1/2/3 degassing under C_2H_4 - Co-olefin pressure: 500 psi — Table 1 - Reaction temperature: 80° C. — Table 1 - Catalyst loading: 0.05 phr — Table 1 - Catalyst loading: 0.10 phr — Table 1 - Catalyst loading: 0.25 phr — Table 1 - Co-catalyst amount: 1.0 phr triphenylphosphine — Table 1 - Cement concentration: 15 wt.-% solid — Example 1/2 - Cement concentration: 6 wt.-% solid — Example 3/4 - Reaction time: 0 to 240 min — Table 2 - Residual double bond content: 100% at 0 min; 0.4% at 240 min — Table 2 (HNBR 1) - Residual double bond content: 100% at 0 min; 3.5% at 240 min — Table 2 (HNBR 2) - Residual double bond content: 100% at 0 min; 0.7% at 240 min — Table 2 (HNBR 3) - Mw: 113371 g/mol — Table 3 (HNBR 1) - Mn: 48352 g/mol — Table 3 (HNBR 1) - PDI: 2.34 — Table 3 (HNBR 1) - Mooney viscosity: 33.3 — Table 3 (HNBR 1) - Mw: 133758 g/mol — Table 3 (HNBR 2) - Mn: 54595 g/mol — Table 3 (HNBR 2) - PDI: 2.45 — Table 3 (HNBR 2) - Mooney viscosity: 41.3 — Table 3 (HNBR 2) - Mw: 195304 g/mol — Table 3 (HNBR 3) - Mn: 64130 g/mol — Table 3 (HNBR 3) - PDI: 3.05 — Table 3 (HNBR 3) - Mooney viscosity: 69.3 — Table 3 (HNBR 3)

Relevant Experimental Evidence - Example 1: Hoveyda-type catalyst gave rapid hydrogenation and low final Mooney viscosity. - Example 2: Arlt catalyst also reduced residual unsaturation and molecular weight, but less effectively than Example 1. - Example 3: Wilkinson's catalyst with triphenylphosphine co-catalyst produced the highest Mooney viscosity among the three HNBR samples. - Example 4: 1-hexene was used as co-olefin, showing the process can be run with a different chain-transfer partner.

Technical Relevance This patent is highly relevant because it provides a direct route to low-molecular-weight HNBR with controlled residual unsaturation. It identifies catalyst system, pressure, temperature, and time as key levers for simultaneously hydrogenating and depolymerizing NBR-derived polymers.

Patent 3

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of NBR in solution using ruthenium, palladium, or rhodium compounds, with the explicit goal of producing HNBR while minimizing catalyst-removal steps and preserving aging properties. The process emphasizes catalyst loading control, hydrogenation temperature, pressure, and time, and it discusses catalyst residue management as a major process issue. - Conjugated diene content: 40 to 90 wt.% — claim 1 - Unsaturated nitrile content: 10 to 60 wt.% — claim 1 - Ruthenium amount: 10 to 200 ppm — claim 1 - Ruthenium preferred amount: 10 to 150 ppm — claim 1 - Ruthenium particularly preferred amount: 10 to 120 ppm — claim 1 - Ruthenium very particularly preferred amount: 10 to 79 ppm — claim 1 - Ruthenium most preferred amount: 42 to 79 ppm — claim 1 - Palladium amount: 20 ppm to <67 ppm — claim 1 - Palladium preferred amount: 44 ppm to <67 ppm — claim 1 - Rhodium amount: 50 ppm to <270 ppm — claim 1 - Rhodium preferred amount: 79 ppm to <270 ppm — claim 1 - Rhodium particularly preferred amount: 50 to 250 ppm — claim 1 - Rhodium very particularly preferred amount: 50 to 170 ppm — claim 1 - Rhodium most preferred amount: 79 to 170 ppm — claim 1 - Hydrogenation temperature: 60 to 200°C — claim 1 - Hydrogenation pressure: 700000 to 15000000 pascals — claim 1 - Hydrogenation time: 1 to 24 hours — claim 1 - Rhodium residue in HNBR: 58 to 60 ppm — cited prior art EP-A-0482391 - Rhodium residue in purified HNBR: 5.8 to 9.2 ppm — cited prior art EP-A-0431370 - Removal rate: 99% — cited prior art CN-A-101704926

Relevant Experimental Evidence - No numbered experimental examples were detected in the provided extraction. - The disclosure emphasizes catalyst residue control as a key quality issue for HNBR aging and appearance. - The patent positions low-catalyst, high-activity hydrogenation as a route to better process economics and product quality.

Technical Relevance This patent is relevant because it defines a practical hydrogenation window for NBR-to-HNBR conversion and explicitly addresses noble-metal catalyst residues. Those residues are important for aging, discoloration, and gel control in HNBR products.

Patent 4

- Patent Number: US5569719A
- Patent Title: Vulcanizable partially hydrogenated nitrile rubber composition containing ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1996
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of partially hydrogenated acrylonitrile-butadiene copolymer rubber by hydrogenating the butadiene

units of NBR in methyl isobutyl ketone using a Pd/SiO₂ catalyst. The resulting HNBR is then compounded with sulfur and tellurium dithiocarbamate vulcanization promoters to obtain vulcanizates with reduced heat build-up and improved mechanical properties. - Bound acrylonitrile content: 36 wt.% — Example 1 - Hydrogenation catalyst: Pd/SiO₂ — Example 1 - Iodine value: 15 — Example 1 - Hydrogenation degree: 95% — Example 1 - Vulcanization temperature: 160° C. — Example 1/2 - Vulcanization time: 20 minutes — Example 1/2 - Sulfur: 0.5 parts — Example 1 recipe a/e - Sulfur: 0.75 parts — Example 1 recipe b/c/d - Sulfur: 1.0 part — Example 1 recipe f - Tetramethylthiuramdisulfide: 1.5 parts — Example 1 - 2-mercaptobenzothiazole: 0.5 parts — Example 1 - Tellurium diethyldithiocarbamate: 1.0 or 2.0 parts — Example 1 - Tellurium diethyldithiocarbamate: 1.0 part — Example 1 recipes c/e/f - Tellurium diethyldithiocarbamate: 2.0 parts — Example 1 recipe d - Iodine value: 10, 20, 30 — Example 2 - Heat build-up: 35, 32, 22, 21, 24, 20 °C — Example 1 table - Dynamic permanent set: 4.5, 3.3, 2.3, 1.9, 3.0, 2.0 % — Example 1 table - Tensile strength: 300, 284, 241, 278, 259, 226 kgf/cm² — Example 2 table - Elongation: 660, 630, 500, 540, 490, 390 % — Example 2 table - Tensile stress at 100% elongation: 82, 89, 107, 113, 126, 144 kgf/cm² — Example 2 table

Relevant Experimental Evidence - Example 1: a 95% hydrogenated NBR was compounded into sulfur-vulcanizable recipes with and without tellurium promoter. - Example 1: tellurium dithiocarbamate-containing recipes showed lower heat build-up and better permanent set than non-tellurium controls. - Example 2: lower iodine values (higher hydrogenation) correlated with lower heat build-up. - Example 2: the tellurium-promoted compositions maintained improved dynamic properties across iodine values 10 to 30.

Technical Relevance This patent is relevant because it links HNBR hydrogenation level to vulcanizate performance and shows how cure chemistry can be tuned for low heat build-up. It provides a concrete example of HNBR precursor hydrogenation and downstream compounding strategy.

Patent 5

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a single-step process in which nitrile rubber is simultaneously hydrogenated and metathesized in the presence of hydrogen, optionally a co-olefin, and a ruthenium- or osmium-based catalyst. The process is intended to produce HNBR with low residual double-bond content, reduced molecular weight, and narrower molecular-weight distribution. - Molecular weight Mw: 20,000 to 250,000 — invention

summary - Mooney viscosity: 1 to 50 — invention summary - MWD/PDI: less than 3.0 — invention summary - Hydrogenation catalyst formula: ruthenium or osmium complex — claim 1 - Hydrogenation/metathesis catalyst ligand system: formula (I) with Y = O, S, N-R1, or P-R1 — claim 1 - Hydrogenation catalyst examples: Grubbs 2nd generation catalyst — cited prior art - Hydrogenation catalyst examples: RuCl₂(PPh₃)₃, RuH(O₂CR)(PPh₃)₃, RuHCl(CO)(PPh₃)₃ — cited prior art - Nitrile rubber Mooney viscosity: 24 to 70 — description - Preferred nitrile rubber Mooney viscosity: 28 to 40 — description

Relevant Experimental Evidence - Example 1: Hoveyda catalyst gave HNBR 1 with the lowest residual double-bond content after 240 min. - Example 1: Arlt catalyst gave HNBR 2 with intermediate hydrogenation performance. - Example 1: Wilkinson's catalyst gave HNBR 3 with the highest Mooney viscosity and highest PDI among the three samples. - Example 1: the inventive catalysts reduced Mw and Mooney viscosity relative to the starting polymer.

Technical Relevance This patent is highly relevant because it directly targets low-molecular-weight HNBR and demonstrates simultaneous hydrogenation/metathesis as a route to processable HNBR. It also provides explicit catalyst-performance comparisons that are useful for selecting synthesis conditions.

Patent 6

- Patent Number: EP2660285B1
- Patent Title: Rubber composition and crosslinked rubber product
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile-rubber compositions based on carboxyl group-containing nitrile rubber and reactive silicone oils, with crosslinking by organic peroxide or polyamine systems. The hydrogenated material is produced by first making a carboxylated NBR latex and then hydrogenating it with a palladium catalyst to obtain a carboxyl group-containing highly saturated nitrile rubber used in the final compositions. - Acrylonitrile: 37.0 parts — Synthesis Example 1 - Mono n-butyl maleate: 4 parts — Synthesis Example 1 - t-Dodecyl mercaptan: 0.5 part — Synthesis Example 1 - 1,3-Butadiene: 57 parts — Synthesis Example 1 - Polymerization temperature: 5°C — Synthesis Example 1 - Cumen hydroperoxide: 0.1 part — Synthesis Example 1 - Polymerization conversion points: 40% and 60% — Synthesis Example 1 - Mono n-butyl maleate addition: 1 part at each of 40% and 60% conversion — Synthesis Example 1 - Polymerization time: 16 hours — Synthesis Example 1 - Hydroquinone aqueous solution: 0.1 part — Synthesis Example 1 - Methanol coagulation: two volumes — Synthesis Example 1 - Vacuum drying: 60°C for 12 hours — Synthesis Example 1 - Palladium content: 1000 ppm — Synthesis Example 1 - Hydrogen pressure: 3 MPa — Synthesis

Example 1 - Hydrogenation temperature: 50°C — Synthesis Example 1 - Hydrogenation time: 6 hours — Synthesis Example 1 - Iodine value of hydrogenated product: 6.5 — Synthesis Example 1 - Mooney viscosity of R1: 12 — Synthesis Example 1 - Mooney viscosity of R2: 40 — Synthesis Example 1 - Carbon black: 200 parts — Example 1 - Side-chain amino-modified silicone oil: 49.5 parts — Example 1 - Polyoxyethylenestearyl ether phosphoric acid: 1 part — Example 1 - 4,4'-di-(α,α -dimethylbenzyl)diphenylamine: 1.5 parts — Example 1 - 1,3-di-o-tolylguanidine: 2 parts — Example 1 - Hexamethylenediamine carbamate: 2.2 parts — Example 1 - Two-end amino-modified silicone oil: 49.5 parts — Example 2 - Two-end epoxy-modified silicone oil: 49.5 parts — Example 5 - Two-end mercapto-modified silicone oil: 49.5 parts — Example 6 - Two-end carboxyl-modified silicone oil: 49.5 parts — Example 7 - Single-end epoxy-modified silicone oil: 49.5 parts — Example 8 - Silica: 50 parts — Example 10 - Tri-2-ethylhexyl trimellitate: 5 parts — Example 10 - 3-aminopropyl triethoxysilane: 1 part — Example 10 - Side-chain amino-modified silicone oil: 5, 10, 15 parts — Examples 10 to 13 - Hexamethylenediamine carbamate: 2.55, 2.51, 2.46 parts — Examples 10 to 13 - DBU: 4 parts — Example 10/11/12/13 - Compound Mooney viscosity: 48, 41, 42, 16, 50, 49, 51, 53, 49, 253, 55 — Table 1/3

Relevant Experimental Evidence - Synthesis Example 1: carboxylated NBR latex was polymerized at 5°C and then hydrogenated at 3 MPa and 50°C to give a highly saturated nitrile rubber with iodine value 6.5. - Example 1: side-chain amino-modified silicone oil produced a composition with good physical properties and oil resistance. - Examples 2 to 7: different reactive silicone oils and crosslinker levels were compared for workability and friction behavior. - Examples 10 to 13: silica-filled formulations showed that reactive silicone oil and silane/crosslinker combinations affect oil resistance, compression set, and skin friction. - Example 15 to 17: reactive silicone oils improved skin friction resistance in sliding-member applications.

Technical Relevance This patent is relevant because it includes an explicit hydrogenation step to make highly saturated nitrile rubber and then demonstrates how that HNBR can be formulated for oil-resistant and low-friction applications. It also provides a clear example of latex hydrogenation followed by coagulation and drying.

Patent 7

- Patent Number: EP4215556A4
- Patent Title: LIQUID HYDROGENATED NITRILE BUTADIENE RUBBER, PROCESS THEREOF AND USE THEREOF
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent concerns liquid hydrogenated nitrile butadiene rubber and a process for making it, but the provided extraction

contains only bibliographic and classification information and no operative synthesis details. Based on the available evidence, the patent is still clearly directed to HNBR production and depolymerization/hydrogenation chemistry, but the specific process parameters are not disclosed in the supplied text.

Relevant Experimental Evidence - No numbered experimental examples were detected in the provided extraction. - The patent is classified under hydrogenation and depolymerisation-related rubber chemistry. - The title and classification indicate a focus on liquid HNBR and its process/use.

Technical Relevance This patent is relevant because it targets liquid HNBR, a lower-molecular-weight hydrogenated NBR derivative that is important for processing and formulation. However, the supplied evidence does not expose the actual synthesis conditions or examples.

Patent 8

- Patent Number: US10465064B2
- Patent Title: Wear resistant and high temperature resistant elastomer nanocomposites
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses elastomer nanocomposites in which a non-functionalized elastomer and a functionalized elastomer are crosslinked with a functionalized filler. HNBR is one of the explicitly listed elastomers, and the disclosure emphasizes crosslinking and filler functionalization rather than the hydrogenation step itself. - Elastomer/filler crosslinking temperature: 25 to 250°C — detailed description - Preferred crosslinking temperature: 50 to 175°C — detailed description - Crosslinking pressure: less than 1 atm to 200 atm — detailed description - Preferred crosslinking pressure: 1 atm to 100 atm — detailed description - Functionalized filler amount: 0.5 to 60 parts per 100 parts elastomer — detailed description - Preferred functionalized filler amount: 0.5 to 30 parts per 100 parts elastomer — detailed description - More preferred functionalized filler amount: 2 to 20 parts per 100 parts elastomer — detailed description - Still more preferred functionalized filler amount: 5 to 15 parts per 100 parts elastomer — detailed description - Elastomer weight ratio (non-functionalized:functionalized): about 200:1 to about 3:1 — claim 2 - Alternative elastomer weight ratio: about 99:1 to about 5:1 — detailed description - Alternative elastomer weight ratio: about 95:5 to about 60:40 — detailed description - Alternative elastomer weight ratio: about 95:5 to about 70:30 — detailed description - Alternative elastomer weight ratio: about 95:5 to about 85:15 — detailed description - Tensile strength at 325°F: about 900 to about 2500 psi — detailed description - Tensile strength at 325°F preferred range: about 900 to about 1600 psi — detailed description

Relevant Experimental Evidence - No numbered experimental examples were detected in the provided extraction. - The patent explicitly includes hydrogenated nitrile butadiene rubber among the usable elastomers. - The disclosure focuses on filler-functionalization and crosslinking for wear and high-temperature resistance.

Technical Relevance This patent is relevant because it places HNBR in a high-performance nanocomposite architecture for wear and high-temperature service. It does not teach HNBR synthesis, but it is useful for downstream formulation and application requirements.

Patent 9

- Patent Number: US10731020B2
- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses rubber compositions for vulcanization molding that use hollow particles and a base rubber, with HNBR listed as one possible base rubber. The process is a compounding and molding disclosure rather than an HNBR synthesis route, but it is relevant to how HNBR can be processed into lightweight molded articles. - Further expansion ratio of hollow particles: 20 to 80% — claim 1 - Hollow particle true specific gravity d1: 0.02 to 0.40 — claim 1 - Base rubber Mooney viscosity: 5 to 90 at 100°C — claim 1 - Base rubber amount: 10 to 95 wt.% — detailed description - Hollow particle amount: 0.1 to 30 wt.% — detailed description - Mooney viscosity of EPDM base rubber: 97 — Comparative Example 1 - Mooney viscosity of EPDM base rubber: 54 — Example 1 - Kneading temperature: 60°C — Example 1/Comparative Example 1 - Kneading time: about 10 minutes — Example 1/Comparative Example 1 - Vulcanization temperature: 140°C — Example 1/Comparative Example 1 - Vulcanization time: 30 minutes — Comparative Example 1 - Molding pressure: 15 MPa — Comparative Example 1 - Vulcanization temperature: 150°C — Example 3/4 - Vulcanization time: 15 minutes — Example 3/4 - Injection cylinder temperature: 180°C — Example 6 - Die temperature: 60°C — Example 6 - Injection pressure: 15 MPa — Example 6

Relevant Experimental Evidence - Example 1: low-breakage hollow particles in an EPDM matrix gave excellent lightweight effect and appearance. - Example 3: extrusion and vulcanization at 150°C produced a lightweight molded product with good dimensional stability. - Example 6: injection molding at 180°C and 15 MPa still preserved lightweight effect and appearance. - The disclosure indicates HNBR can be used as a base rubber in the same lightweight molding framework.

Technical Relevance This patent is relevant to HNBR because it identifies HNBR as a compatible base rubber for lightweight molded products. It is not a synthesis patent, but it informs downstream processing constraints such as Mooney viscosity and molding temperature.

Patent 10

- Patent Number: US11131058B2
- Patent Title: Method for manufacturing core wire for transmission belt, treatment agent, and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a cord-treatment method for power transmission belts using a first treatment agent containing resorcin-formaldehyde condensate, carboxy-modified latex, and polycarbodiimide resin in water. Carboxy-modified hydrogenated NBR latex is explicitly listed as one of the latex options, making the patent relevant to HNBR-derived adhesive systems rather than HNBR synthesis itself. - Condensate (A1) proportion: 10 to 100 parts by mass per 100 parts by mass of rubber component (A2) — claim 1 - Polycarbodiimide resin NCN equivalent: 300 to 450 — claim 1 - Curing agent (A3) proportion: 1 to 10 parts by mass per 100 parts by mass of rubber component (A2) — claim 2 - Rubber composition (A) proportion: 15 to 20 mass% of first treatment agent — detailed description - First treatment agent immersion time: 10 seconds — Adhesion treatment - Drying temperature: 60°C — Adhesion treatment - First treatment agent total solid content concentration: 17 to 20% by mass — Table 1 - Second treatment agent total solid content concentration: 5% by mass — Table 4 - Third treatment agent composition: EPDM 100 parts, stearic acid 5 parts, zinc oxide 5 parts, HAF carbon black 10 parts, hydrous silica 15 parts, RF condensate 2 parts, anti-oxidant 2 parts, MBTS 2 parts, hexamethoxy methylol melamine 2 parts, sulfur 1 part — Table 5 - Third treatment agent composition: EPDM blending rubber composition 93 parts, polymeric isocyanate 7 parts, toluene 900 parts — Table 6

Relevant Experimental Evidence - Example 5: RFL-X with COOH-modified NBR and polycarbodiimide dispersion B was prepared at 18 to 19% solid content. - Example 9: alternative RFL systems using COOH-modified NBR or COOH-modified VP were prepared with different polycarbodiimide loadings. - The patent demonstrates that carboxy-modified hydrogenated NBR latex is a viable adhesive latex option for belt cords. - The process emphasizes water-based treatment and micelle-forming polycarbodiimide resin for improved adhesion.

Technical Relevance This patent is relevant because it explicitly includes carboxy-modified hydrogenated NBR latex in a water-based adhesion system. While not a synthesis route, it shows a downstream use case for HNBR-derived latex materials in belt reinforcement.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US4486480A and US5569719A both emphasize partially hydrogenated NBR with high diene saturation, but US4486480A focuses on fuel-hose durability while US5569719A focuses on vulcanizate heat build-up and cure promotion.
- US7951875B2 and US9150669B2 both target low-molecular-weight HNBR, but US7951875B2 uses simultaneous hydrogenation/metathesis with explicit catalyst loading and co-olefin conditions, whereas US9150669B2 emphasizes alternative Ru/Os catalyst systems and reports lower Mw/PDI outcomes.
- EP3827029A1 is the most catalyst-residue-focused hydrogenation patent, specifying Ru/Pd/Rh ppm windows and broad temperature/pressure/time ranges, while US4486480A and US5569719A are more application/formulation oriented.
- US5569719A and EP2660285B1 both connect hydrogenated nitrile rubber to compounding performance; US5569719A uses sulfur/tellurium cure chemistry, whereas EP2660285B1 uses reactive silicone oils, polyamine/peroxide crosslinking, and a prior hydrogenation step with Pd catalyst.
- EP2660285B1 and US11131058B2 both involve carboxylated nitrile systems, but EP2660285B1 uses carboxylated highly saturated nitrile rubber as a final elastomer, while US11131058B2 uses carboxy-modified hydrogenated NBR latex as an adhesive component.
- US10465064B2 and US10731020B2 are downstream application patents rather than synthesis patents; both include HNBR as a usable elastomer, but US10465064B2 targets nanocomposite wear/high-temperature performance and US10731020B2 targets lightweight molded products with hollow particles.
- EP4215556A4 is the least experimentally detailed of the set in the provided evidence, but its title indicates liquid HNBR and likely overlaps with the low-Mw direction seen in US7951875B2 and US9150669B2.
- Across the set, the recurring synthesis levers are hydrogenation degree, catalyst identity/loading, solvent-phase processing, and post-hydrogenation compounding to balance residual unsaturation, cure response, and aging resistance.

4. CONCLUSION

The evidence shows that hydrogenated NBR synthesis is best controlled by managing hydrogenation degree, catalyst system, and molecular-weight retention. For direct HNBR production, the strongest process evidence supports solution-phase hydrogenation with noble-metal catalysts at roughly 60 to 200°C and elevated hydrogen pressure, while low-molecular-weight HNBR can be obtained by combining hydrogenation with metathesis using Ru/Os catalysts and carefully tuned catalyst loading. For application performance, the most useful HNBR targets are high diene saturation with residual unsaturation low enough to improve heat,

ozone, and oil resistance but not so low that cure becomes difficult; the patents suggest that partial hydrogenation levels around 50% or higher are a minimum practical threshold, with much higher hydrogenation often preferred. Recommended evidence-based synthesis parameters are: solution hydrogenation, noble-metal catalyst ppm in the tens to low hundreds, hydrogen pressure in the MPa range, temperature around 50 to 200°C depending on catalyst, and reaction times from 1 to 24 hours. Downstream, sulfur/tellurium cure systems, reactive silicone oils, and functionalized fillers are effective routes to convert HNBR into durable hoses, seals, and nanocomposites.

5. REFERENCES

- US4486480A | Rubber hose comprising a liquid nitrile-butadiene copolymer incorporated into a ... | Not disclosed | US | 1984 | <https://patents.google.com/patent/US4486480A/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- US5569719A | Vulcanizable partially hydrogenated nitrile rubber composition containing ... | Not disclosed | US | 1996 | <https://patents.google.com/patent/US5569719A/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- EP2660285B1 | Rubber composition and crosslinked rubber product | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP2660285B1/en>
- EP4215556A4 | LIQUID HYDROGENATED NITRILE BUTADIENE RUBBER, PROCESS THEREOF AND USE THEREOF | Zannan Scitech Co Ltd | EP | 2024 | <https://patents.google.com/patent/EP4215556A4/en>
- US10465064B2 | Wear resistant and high temperature resistant elastomer nanocomposites | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10465064B2/en>
- US10731020B2 | Rubber composition for vulcanization molding, process for manufacturing the ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731020B2/en>
- US11131058B2 | Method for manufacturing core wire for transmission belt, treatment agent, and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11131058B2/en>